Conditional Probability

Basic probability rules

1 A bag contains 5 red marbles, 3 blue marbles, and 2 green marbles. Find the probability of drawing a blue marble.

2 A fair six-sided die is rolled. Find the probability of rolling an even number or a 5.

Two-way tables and conditional probability

3 A survey of 300 people found: 90 own a dog and a cat, 60 own only a dog, 40 own only a cat, and 110 own neither. Find the probability a randomly selected person owns a cat, given that they own a dog.

4 Using the same survey data, find the probability a randomly selected person owns a dog, given that they own a cat.

Independent vs. dependent events

5 Explain the difference between independent and dependent events, and classify the following: drawing two cards from a deck without replacement.

6 A jar has 4 red and 6 blue chips. Find the probability of drawing two red chips in a row, without replacement.

The multiplication rule for independent events

7 A fair coin is flipped and a fair six-sided die is rolled. Find the probability of getting heads and rolling a 4.

Probability with replacement vs. without replacement

8 A deck of 10 cards numbered 1-10 has one card drawn, replaced, then a second card drawn. Find the probability both cards drawn are greater than 7.

9 Using the same deck of 10 cards but WITHOUT replacement this time, find the probability both cards drawn are greater than 7.

Interpreting conditional probability in real-world contexts

10 A medical test is 95% accurate for detecting a disease that affects 1% of the population. Explain, in words, why a positive test result does not necessarily mean a 95% chance of having the disease, referencing the low overall disease rate.

The complement rule

11 A spinner has 8 equal sections numbered 1-8. Find the probability of NOT landing on a number greater than 6, using the complement rule.

Probability trees for multi-stage events

12 A box has 3 red and 2 white balls. Two balls are drawn without replacement. Find the probability that the first is red and the second is white.

13 Using the same box (3 red, 2 white, no replacement), find the probability of drawing two balls of the same color.

Comparing theoretical and experimental probability

14 A fair coin is flipped 50 times and lands heads 32 times. Compare the experimental probability of heads to the theoretical probability, and explain a possible reason for the difference.

Combining conditional probability with two-way tables

15 A two-way table of 400 students shows: 180 play an instrument and are in band; 20 play an instrument but are not in band; 40 don't play an instrument but are in band; 160 neither play an instrument nor are in band. Find the probability a student is in band, given that they play an instrument.

16 Using the same table (180 instrument+band, 20 instrument only, 40 band only, 160 neither, total 400), find the probability that a randomly selected student plays an instrument OR is in band (not both counted twice).

A multi-part conditional probability problem

17 This question has three parts. A school has 600 students: 350 take Spanish, 200 take French, and 50 take both languages. (a) Find the probability a randomly selected student takes at least one of the two languages. (b) Find the probability a student takes Spanish, given that they take at least one language. (c) Find the probability a student takes neither language.